



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-301-1

Date: 19-7-2022

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TECHNICAL SPECIFICATION FOR LIVE WORKING –
POTABLE EQUIPMENT FOR EARTHING
MEDIUM /LOW VOLTAGE NETWORKS

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1- SCOPE

This specification covers the minimum technical requirements for design, engineering, manufacturing, testing live working portable equipment for earthing medium/low voltage networks. The main reason for their installation is to protect the worker from accidental power on or return of voltage during repair work.

2- APPLICABLE STANDARDS

Unless otherwise specified in this specification, live working – portable equipment for earthing medium/low voltage networks should comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance with the applicable IEC standards as following:

S. No	Standard No.	Description
1	IEC61230	Live working – Portable equipment for earthing or earthing and short-circuiting
٢	IEC60228	Conductors for Insulated cables
3	IEC61235	Live working - Insulating hollow tubes for electrical purposes



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3. CONSTRUCTION REQUIREMENTS

- Cables shall have adequate mechanical, chemical, environmental, and electrical properties to meet all requirements of IEC61230 standard.
- The cable conductors should have standard nominal cross-sectional area according to IEC 60228.
- Cables for earthing and short-circuiting purpose shall be flexible and insulated for mechanical protection.
- The design of the portable earthing equipment should be made to fulfill short circuit current conditions in the sold earth networks.
- The connection between end fittings and cables shall be protected from water penetration.
- Clamps shall provide reliable contact performance and shall withstand the thermal and mechanical stresses produced by the rated short-circuit current.
- Line and earth clamp shall be suitable for the surface and shape of the connecting point and shall permit easy and safe installation without causing damage to clamps and without danger for personnel
- The insulation material shall be made to fulfill chemical, environmental and temperature conditions met in practice.
- The insulating pole shall be shortened for transport by telescoping or by disassembly

4. TYPES OF EARTHING EQUIPMENT

4.1. Live working – portable equipment for earthing medium voltage overhead lines

- Three aluminum alloy clamps for cylindrical conductors, flat bars
- One dispenser head with hook
- Three copper cables with transparence pvc insulation of 35 mm² section and 2.5 m length
- One copper cables with transparence pvc insulation of 35 mm² section and 12 m length
- One metallic spool to store the 12 m of earthing cable.
- One insulating pole (1.25m folded and 2m extended)
- One earthing rod



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4.2. Live working – portable equipment for earthing medium voltage switch board and transformers

- Three aluminum alloy clamps for cylindrical conductors, flat bars
- Three copper cables with transparence pvc insulation of 35 mm² section and 0.75m length
- One copper cables with transparence pvc insulation of 35 mm² section and 2m length
- One insulated connecting cluster
- One earthing rod
- One earthing lathe
- One insulating pole ≥ 1.25 m length.

4.3. Live working – portable equipment for earthing low voltage overhead lines

- Five aluminum alloy clamps with 50-70cm pole for cylindrical conductors
- Four copper cables with transparence pvc insulation of 25 mm² section and 0.6 m length
- One copper cable with transparence pvc insulation of 25 mm² section and 12m length
- One earthing lathe
- One earthing rod

4.4. Live working – portable equipment for earthing low voltage electric panels

- Four aluminum alloy clamps with for cylindrical conductors, flat bars
- Four copper cables with transparence pvc insulation of 25mm² section and 0.6 m length
- One copper cable with transparence pvc insulation of 25mm² section and 2 m length
- One earthing lathe
- One earthing rod

5-MARKING

The marking should be durable and easily legible. Each of the following component of live working – portable equipment for earthing medium/ low voltage networks has name plate which should carry at least the following items of marking. Compliance with the following markings items will be checked by inspection.



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5-1 Marking on earthing and short circuiting device

- Manufacturer's name or trade mark or end assembler's name
- Type/Model no.
- Year of manufacture.
- Rated current I_r (KA), rated time T_r (s) of the device and value of peak factor
- Number of the relevant IEC standard.

5-2 Marking on clamp

- Manufacture's name or trade mark.
- Model or type reference for the clamp.

5-3 Marking on cable

- Indication of the origin of the cable.
- Cross-section or cable size.
- Nature of the insulating covering.

6-TESTS

Type Tests of portable equipment for earthing medium/low voltage Networks

➤ According to IEC 61230

1. Fatigue and humidity penetration tests on cable with end fittings
2. Tension test on cable with end fittings
3. Test on clamps ,permanent connection point and connections
4. Short circuit current test

➤ Telescopic stick according to IEC 61235.

1. DRY/WET power frequency voltage withstand test
2. Crushing test
3. Bending test
4. Dye penetration test.



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7- SAMPLING TESTS

The supplier shall test 10% from supply batch order which taken randomly and witnessed by Distribution Company representative. The supplier shall pay all costs of tests.

8-DRAWING AND CATALOGUES:

- All necessary drawings including layout, dimensions should be submitted with the offer.
- Technical catalogues which contain all technical data should be submitted.
- Installation and maintenance manual should be delivered after acceptance.

9-GUARANTEE

The supplier should guarantee the Type Tests of portable equipment for earthing medium/low voltage Networks against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier.

- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge inside guarantee period.

9.1- Guarantee table

- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.



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**Guarantee Table For
M.V /L.V Voltage Detector Range from 50V TO 36 kV**

ITEM	Guarantee	Supplier
1. <u>Manufacturer's name</u>		
2. <u>Standard specifications applied</u>	IEC 61230 IEC61235 IEC 60228	
3. <u>Earth cable</u> -material -cross section -length		
4. <u>Short circuit cable</u> -material -cross section -length		
5. <u>Earthing clamp</u> -material -type		
6. <u>Line clamp</u> -material -type		
7. <u>Power specification</u> -rated current -max. peak current, time		
8. <u>Environmental specifications</u> -operating temperature -humidity -storage temperature		
9. <u>Insulated pole</u> -material -length of insulation element -length of handle -overall length		
10. <u>Earth rod</u> -material -length		

We guarantee the data given above for the equipment offered.

Signature: Date: